

BRIGHT WIREDU NUA KOH

Email: bryteredunuakoh@gmail.com | **Mobile:** +358 41 7244264 | **LinkedIn:** [Bright Wiredu](#)
GitHub: [Wiredu2020](#) | **Website:** [Bright portfolio](#)

EDUCATION

Master of Science – Data-Centric Engineering

August 2024 - July 2025

Lappeenranta-Lahti University of Technology (LUT), Finland

Thesis: 3D Reconstruction of Wild Animals for Re-identification

Supervisor: Assoc. Prof Tuomas Eerola, LUT University

Relevant Courses: *Advance Data Analysis and Machine Learning, Digital imaging and Image Preprocessing, Pattern Recognition and Machine Learning, 3D Vision, Machine Vision and Data analysis*

Master of Science – Mathematical Sciences

August 2023 - June 2024

African Institute for Mathematical Sciences (AIMS), Rwanda

Thesis: Super-Resolution Methods for Satellite Images Application

Supervisor: Dr. D.Sc. Abebe Geletu, German Researcher-AIMS

Relevant Courses: *Statistical Machine Learning, Introduction to Neural Network, Interpretable Machine Learning, Data-Driven Optimization, Big Data Analytics with Python, Statistical Regression, Stochastic Processes, Algorithms and Data Structures, Differential Equations (Partial and Ordinary), Discrete Mathematics, Numerical Linear Algebra, Actuarial Mathematics, Financial Mathematics.*

Bachelor of Science – Mathematics with Computer Science

August 2018 – Nov. 2022

University of Ghana, Accra, Legon

RESEARCH EXPERIENCE

Junior Research Assistant: Training Data Preparation Noguchi Ghana

November 2022 – Feb 2023

- Researched methods for handling real-time data streams, including preprocessing, reduction, and integration.
- Experimented with active learning techniques to continuously update models with new data while ensuring stability and efficiency.

Super-Resolution Methods for Satellite Image Application

March – June 2024

- Trained SRCNN, EDSR and SRGAN to reconstruct satellite images with SRGAN achieving the highest average accuracy of 0.83 SSIM and 35.5 dB (PSNR) on test set.

THESIS PUBLICATION

3D Reconstruction of Wild Animals for Re-identification [LUT Pub](#)

PAPER UNDER REVIEW

Afrifa-Yamoah et al., *Predictive spatial machine learning in the era of big data: a critical analysis of methods, technical advances, and research frontiers*, Submitted 2025

UNIVERSITY PROJECTS

- Robot Navigation and Object Detection with Computer Vision [Project Link](#)
- Glioblastoma Tumor Segmentation Using Deep Learning [Project Link](#)
- BBC News Classification and Text Generation with Transformer Model [Project Link](#)
- Cancer Classification in Medical Data using RNN and LSTM [Project Link](#)
- Image Level Micro-Gesture Classification with CLIP [Project Link](#)
- Understanding Graph Neural Network [Project Link](#)
- Coin Detection and Estimation from Images [Project Link](#)
- Anomaly Detection using Deep Learning [Project Link](#)
- Mono Sound Wave Filtering [Project Link](#)

- Physics-Informed Neural Network for Heat Equation [Project Link](#)

CERTIFICATIONS

- ICDL Professional Data Analytics

TECHNICAL SKILLS

- **Programming Languages:** Python, R, MATLAB, SQL, Java, C++
- **Image & Signal Processing:** OpenCV, scikit-image, SciPy, Pillow, PyWavelets, NumPy
- **ML and Deep Learning:** Pytorch, Tensorflow, Streamlit, Keras, PySpark
- **Web Development:** Django, HTML, CSS, React, Node.js, RESTful APIs
- **Data Analysis and Visualization:** Power BI, Matplotlib, R, MATLAB
- **Version Control and Collaboration:** GitHub, Unix Shell
- **Big Data Technologies:** Apache Spark, PostgreSQL, MySQL

TEACHING EXPERIENCE

Assistant Lecturer: AIMS Rwanda

Aug 2025 – Current

- Assisted faculty in teaching deep learning and programming
- Discharged administrative duties such as grading assignments and managing students' records.

Teaching Assistant: University of Ghana, Department of Mathematics

Nov 2022 - Aug 2023

- Assisted students with tutorials that helped reinforce the concepts taught in class.
- Mentored students who were passionate about coding and programming.
- Discharged administrative duties like attendance taking, grading assignments, and managing students' records.

PROFESSIONAL EXPERIENCE

Junior ML Engineer for Annologic LLC, Ghana Accra

July 2025 – Current

- Applied AI models on data cleaning task within ETL pipelines on AWS
- Built scraping app using Selenium and FastAPI for housing website

AWARDS AND PROFESSIONAL AFFILIATION

- Lappeenranta University of Technology (LUT) Scholarship of Merit, Finland, 2024 -2025
- African Institute for Mathematical Sciences Scholarship of Merit, Rwanda, 2023 - 2024

REFEREES

Prof. Blaise TCHAPNDA

- AIMS Rwanda, Academic Director
- E-mail: blaise.tchapnda@aims.ac.rw

Dr. D.Sc. Abebe Geletu

- Master's Thesis Supervisor
- German Research Chair at AIMS Rwanda
- E-mail: abebe.geletu@aims.ac.rw

Dr. Joseph K. Ansong

- Bachelor's Project Supervisor
- Head of Mathematics Department, University of Ghana
- E-mail: jkansong@ug.ed.gh